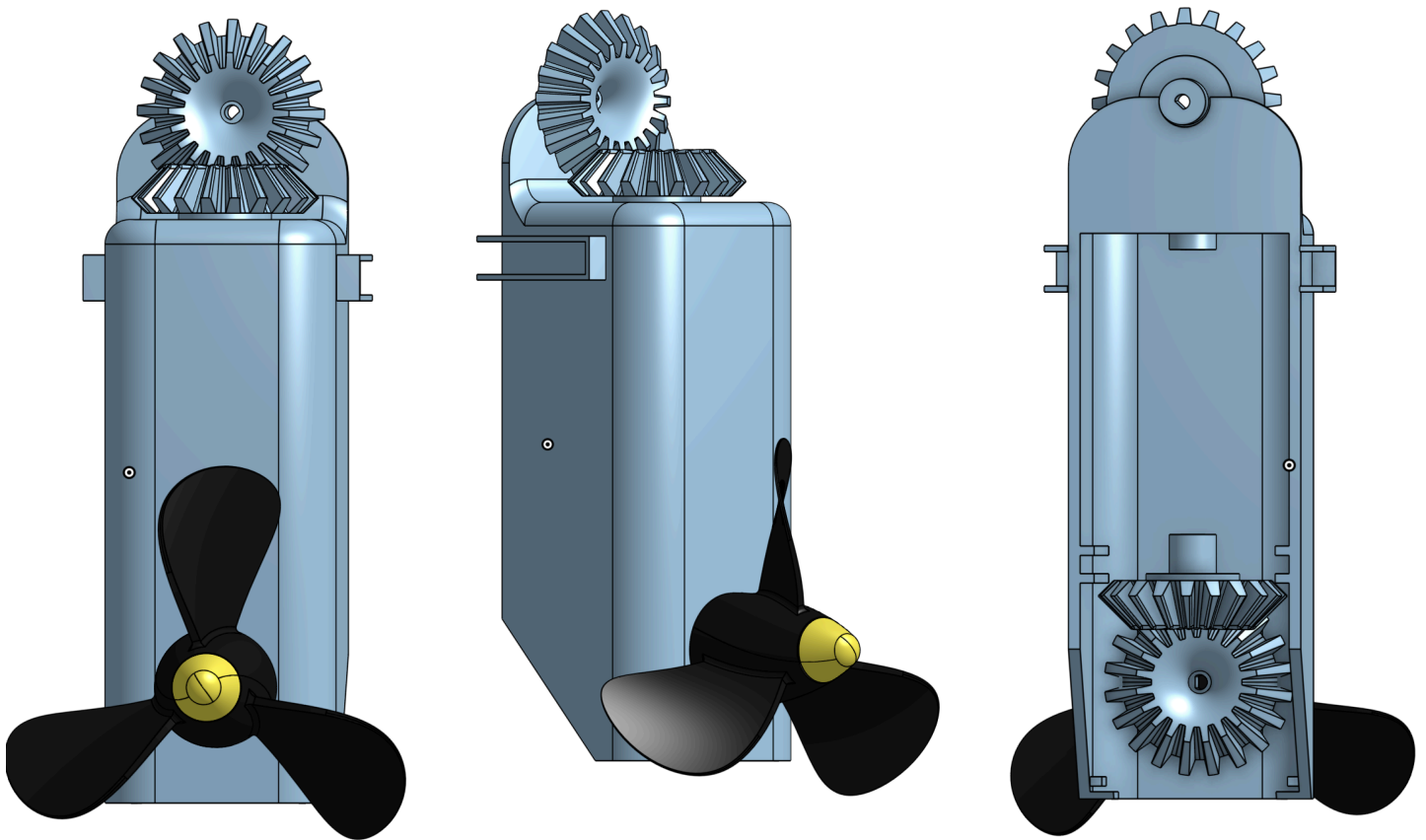


Motor

The system utilizes a four-gear gearbox designed to redirect the rotational motion from a horizontally mounted motor to a horizontally aligned propeller, while temporarily converting the motion into a vertical orientation. This allows for a compact and efficient transfer of mechanical energy within the hull of the boat.

The fourth gear is aligned horizontally and meshes with the third gear, maintaining the final direction of motion. Attached to the shaft of the fourth gear is a propeller, which converts the rotational motion into thrust. The propeller's rotation pushes water backward, propelling the boat forward.



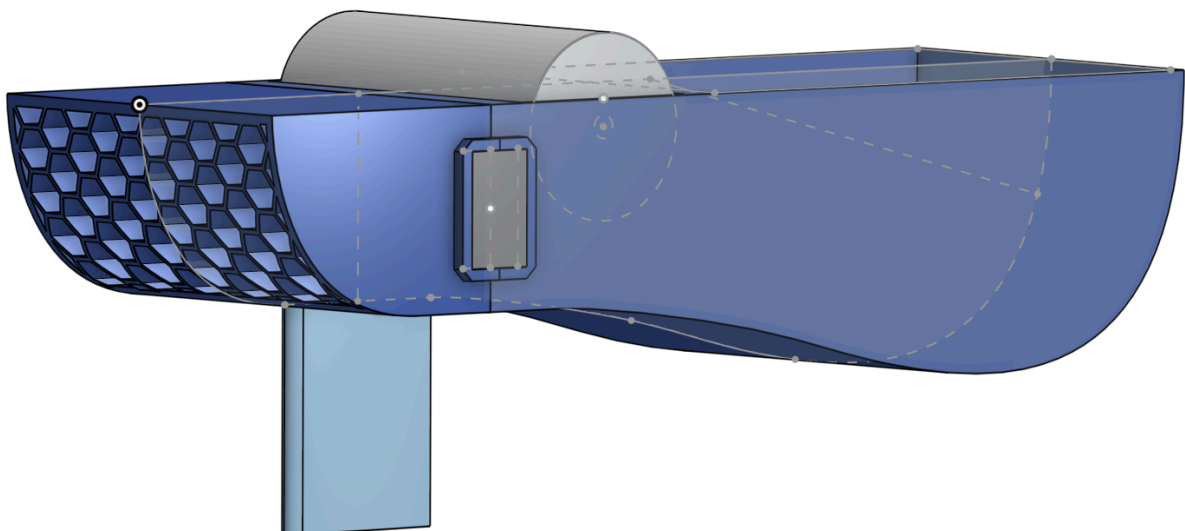
Oil basin and filter

As the boat moves across the surface of the water, a collection intake located at the front or base of the hull draws in a mixture of oil and seawater. This intake is designed to skim the surface layer, where oil tends to accumulate due to its lower density compared to water.

The collected liquid mixture enters a sealed internal chamber where the separation process takes place. Inside this chamber is a specially designed hydrophobic (water-repellent). Oil adheres to the hydrophobic sheet, which attracts and traps the oil. Water is repelled by the sheet and is drained out of the system back into the ocean.

The absorbed oil is funneled from the hydrophobic sheet into an onboard container. This container is sealed to prevent leakage and protect the recovered oil from mixing with seawater again. Sensors or transparent panels may be used to monitor the oil level within the storage tank.

Once the oil collection container is full or the cleanup operation concludes, the oil is transported to a station where it can be safely managed, recycled, or disposed of according to environmental protocols and local regulations.

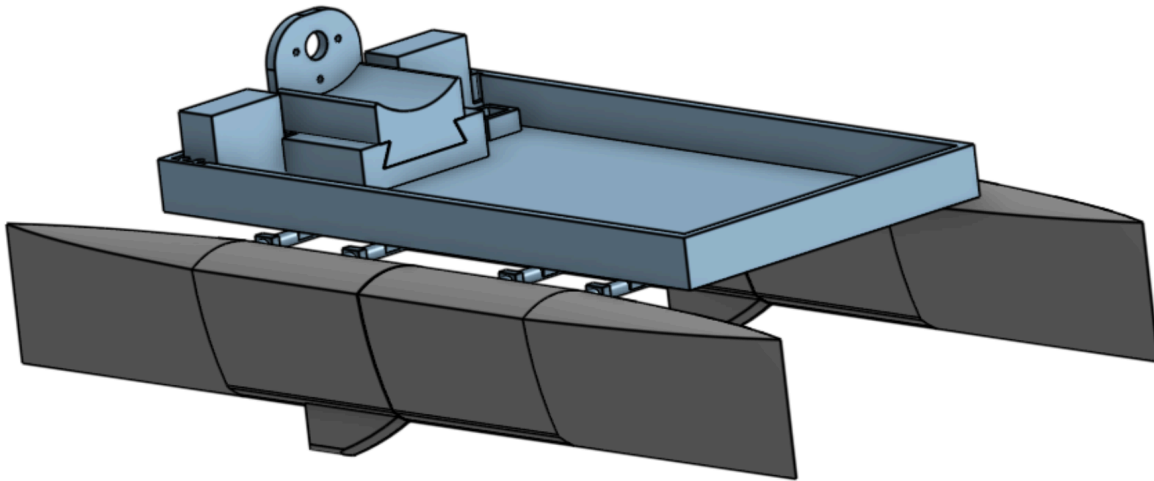


Boat

The boat uses a catamaran configuration, consisting of two parallel hulls connected by a central platform. This design offers:

- Excellent stability, even in choppy water
- Wide deck space for equipment and storage
- A natural channel between the hulls, ideal for directing surface water and oil into the collection system

The space between the hulls is strategically used to guide the oil-water mixture into the boat's intake chamber, ensuring maximum surface coverage during cleanup operations.



Our Story – Team ORCA

We are the **Oil Recovery and Collection Association**, a group of high school students from the UBC Geering Up STEM Leadership Program. We wanted to tackle a real-world problem and decided to choose oil spills, due to how much they affect our planet and ecosystem. We designed and built a working **oil skimmer** prototype to recover large amounts of oil from future spills. Research shows that on average, **only around 25% of spilled oil is ever recovered**, leaving most of it to sink into the ocean and severely damage ecosystems.

Our Mission

We believe in a future where oil spills no longer pollute the oceans and coasts. Our mission is to make oil recovery **fast, affordable, and accessible worldwide**. By protecting ecosystems and wildlife, we also protect our food and communities. ORCA is committed to building a safer, cleaner future for our planet.

Why Oil Spills Matter

Oil spills happen when crude oil leaks from tankers, drilling rigs, or offshore wells. These spills spread quickly, creating a toxic layer that affects marine ecosystems. The fish, birds, and mammals can be poisoned, lose their habitats, or die from exposure. The effects of oil can last for decades, harming both the environment and us.

Our Solution – SpillBot

To solve this problem, we created **SpillBot**, a small and effective oil recovery boat/tow. SpillBot is affordable and can be easily **3D printed**, making it easy to assemble anywhere in the world. It is designed to be **fast, reliable, and efficient** at removing oil from the water's surface. Our vision is to partner with governments, coast guards, and shipping companies so that SpillBot can be deployed whenever any spills occur.

What Happens to the Oil?

Once collected, the oil goes under the process of **bioremediation**, where natural microorganisms or plants break down the oil into harmless substances. If the oil is too impure it may be carefully **burned off** as a last resort.

Our Process

- After having decided on exploring the problem of oil spills and how to find a solution, our project began with brainstorming the different ways we could collect spilled oil from the ocean.
- We researched the problem of oil spills and how current day technology is used to collect spilled oil
- We initially came up with 4 solutions, and one by one we listed the pros and cons to each model.
- Our two final ideas came down to the skimmer design and the oil filtration design.
- We decided on using the skimmer as it is easy to replicate and it is a unique way to collect oil.
- We then split into two groups, with one group focusing on designing the skimmer, and one group focusing on designing the boat.
- To begin our designing process, we made multiple online sketches
- We then used CAD (computer aided design), which allowed us to add specific details, and eventually 3D print all of the different pieces of the boat.
- These pieces were then connected and glued together to form the robot.

